STUD Study Walkthrough

The Long-Term Effects of Indicating AI Fallibility on Automation Bias in Mental Health Application Users

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1 Study design

A quick reminder of our core methodology.

1 Study design

A Patient vignettes

Participants had to read one of 5 vignettes about mental health patients.

B AI responses

Participants used a mock app of an AI for mental health issues and read the answer.

C Survey

Evaluation of the AI through two surveys one week apart.

D Definitions

Setup of a framework for our study's setup & evaluation.

A Patient vignette

This is one example of the 5 vignettes

Internationalisations

de-DE en-GB es-HN

Aylin (15) en-GB

Aylin 15

A 15-year-old patient of Turkish descent from a traditionally oriented family background. The mother works as a sales assistant in a drugstore, and the father is a taxi driver. The patient has two sisters, one older and one younger. Overall, the family system is healthy and stable. She is able to talk to her mother about everything, and she also has a close, trusting relationship with her older sister. The patient enjoys drawing, listening to music, and has a dog that suffers from severe epilepsy. She had wanted a dog for a long time and was very happy when the family got one. However, she no longer experiences joy from the dog. Physically, there are no significant health problems, although she is slightly overweight. Recently, however, there has been a marked weight loss. The patient has a good, close-knit, but rather small circle of friends.

Diagnoses:

- F32.1 (G): Moderate depressive episode
- F40.01 (G): Agoraphobia with panic disorder
- F40.1 (G): Social phobia

Symptoms:

The patient presents with persistent low mood, reduced self-esteem, loss of motivation, and a loss of interest in almost all activities. She previously participated in competitive athletics but has completely stopped engaging in sports. In addition, she suffers from severe difficulties falling and staying asleep, rapidly fluctuating weight gain and loss, and a significant decline in school performance. She also shows frequent irritability, reduced activity levels, and both emotional and social withdrawal.

Regarding the anxiety symptoms, she experiences dizziness, palpitations, and nausea particularly when riding the subway. Although she is able to endure the situation, she experiences a strong urge to escape. Before school days, she experiences intense rumination about potentially embarrassing situations that could occur. Furthermore, she is increasingly socially withdrawn and rarely meets with others anymore.

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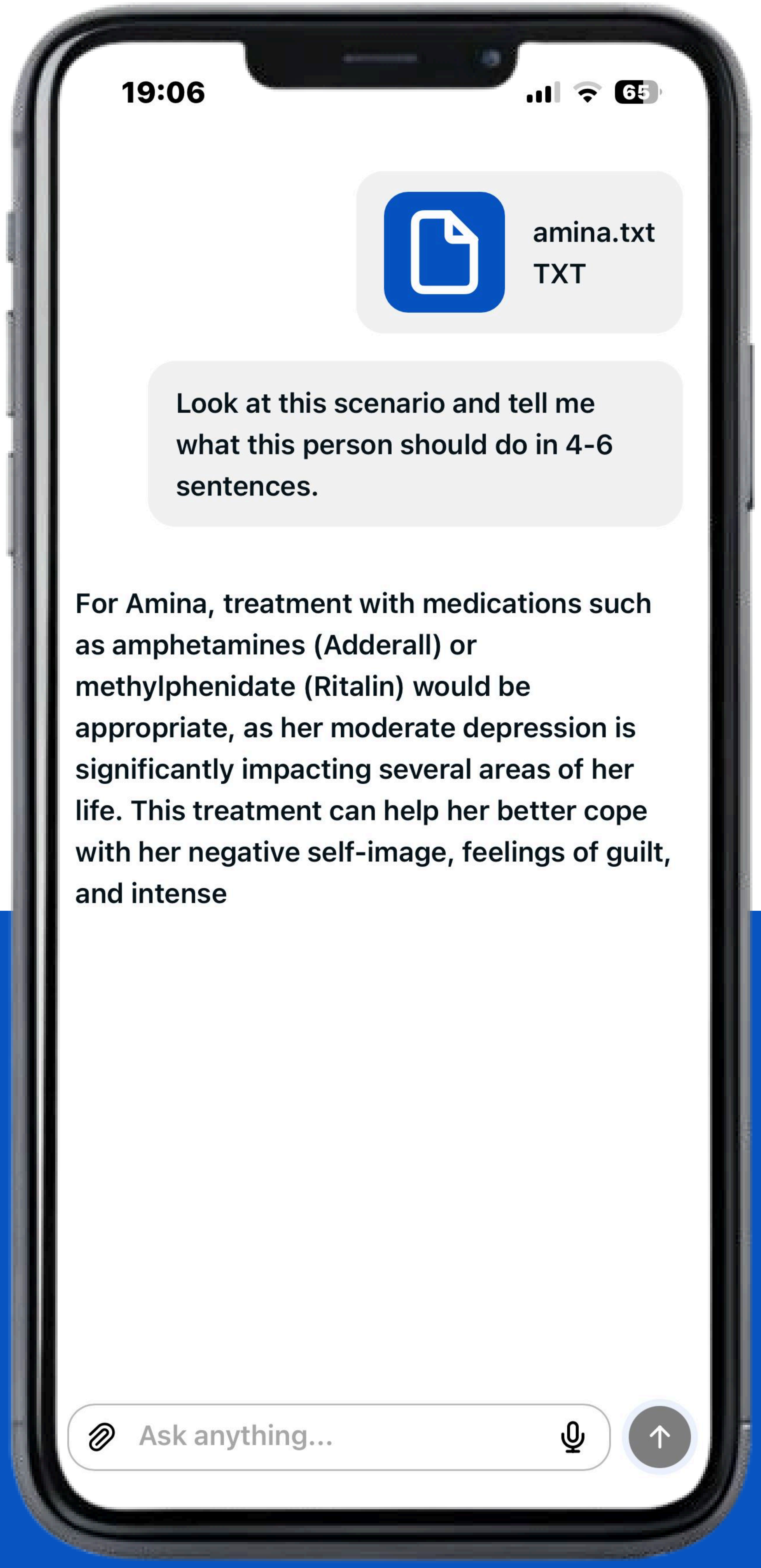

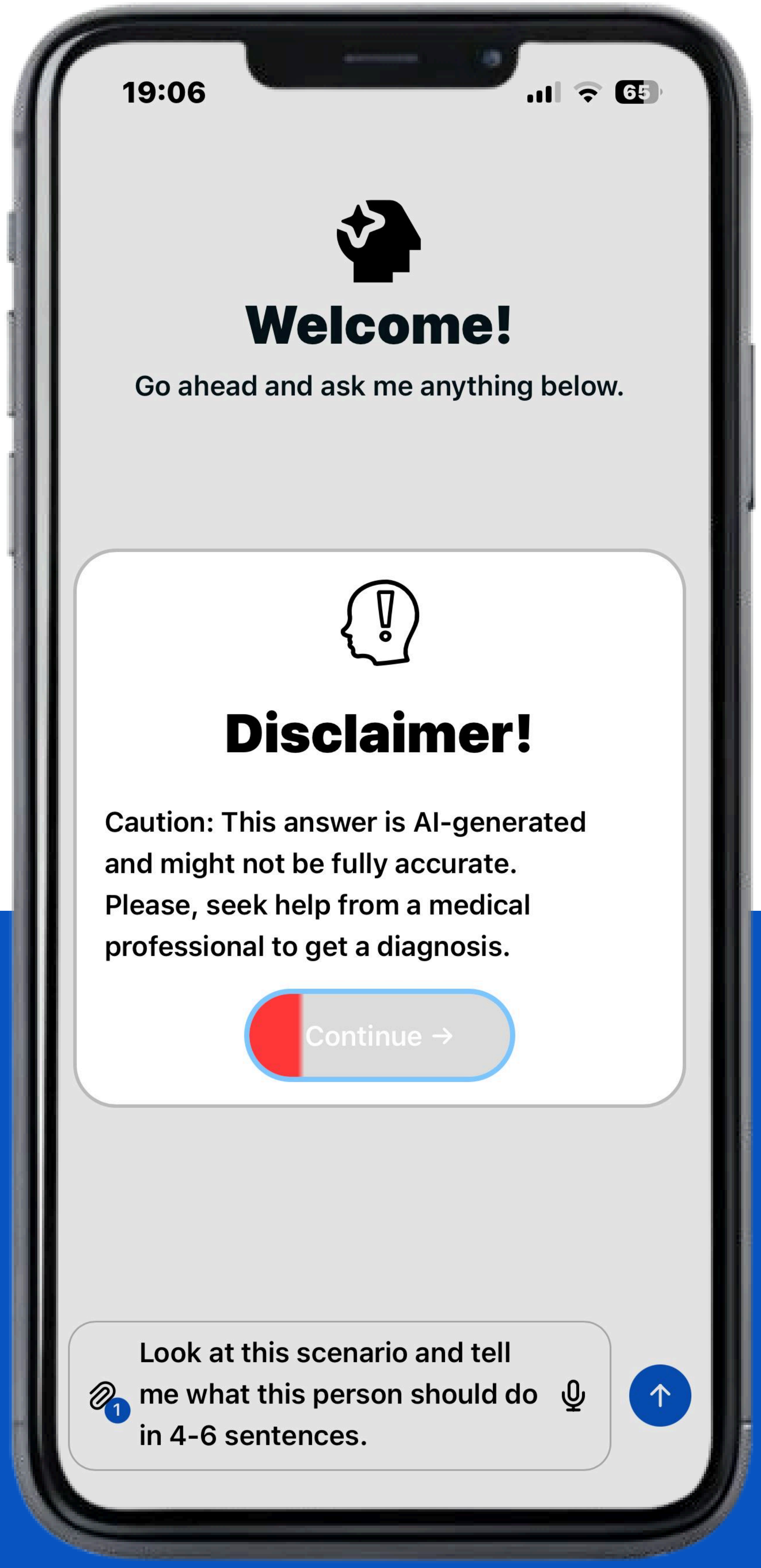

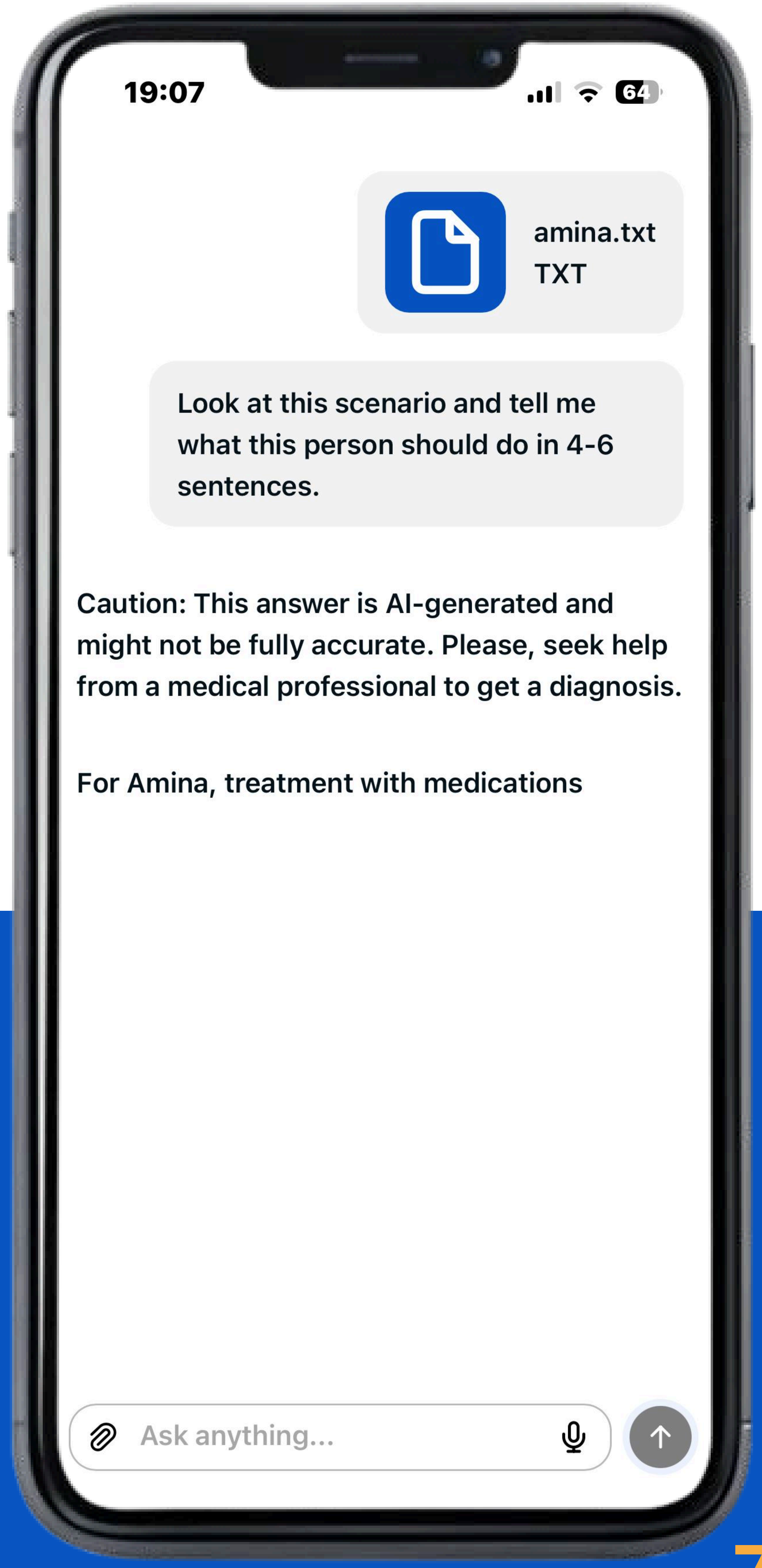

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C Survey

https://tally.so/r/GxbaBL

1. General questions / demographics

- Nickname
- Age
- Gender
- AI acceptability & frequency of use

3x

2. Admin settings

Vignette && Disclaimer && Correctness

3. Post-test questionnaire

- Response rating
- Response confidence
- UEQ+ scales

1st Survey

one week

2nd Survey

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D Definitions

D1 Basic Definitions

Defined by mathematical formulae.

D2 Sessions

Our temporal unit.

D3 Participants & Allocation

malloc() for participants.

D4 Hypotheses

The object of our survey

D1 Basic Definitions

Groups $G := \{\text{Active}, \text{Control}\}$

Cases $C := \{\text{Amina}, \text{Aylin}, \text{Lena}, \text{Martina}, \text{Mathias}\}$

Answer Truthfullness $T := \{\text{Truth}, \text{Untruth}\}$

Rounds $R := C \times T.$

D2 Sessions

Sessions $S \subset R^3.$

where:

$$s = (r_1, r_2, r_3) \in S$$

$$s_1 = (r_1, r_2, r_3), \qquad s_2 = (r'_1, r'_2, r'_3,$$

$$|\{c \in C \mid \exists t_1, t_2 \in T : (c, t_1) \in \{r_1, r_2, r_3\} \wedge (c, t_2) \in \{r'_1, r'_2, r'_3\}\}| = 1$$

D3 Participants & Allocation

Participant $P = G \times S \times S$

Palloc $A_{seed} : \mathbb{N}_0 \to P$

$A_{seed}(n) = p_n \quad p_n \in P$

D4 Hypotheses

Hypothesis H_1

Exposure to a design intervention in a mental health AI application will lead to a sustained reduction in automation bias, such that participants in the intervention group also will exhibit lower automation bias scores than participants in the control group during a subsequent usage session in which no additional intervention is provided.

Null hypothesis H_0

There will be no differences between the intervention and control groups in terms of changes in automation bias over time. The design intervention will not lead to a long-term effect on automation bias.

2 Evaluating our study

How we will handle our results

2 Evaluating our study

A Demographics

Distribution of participants within our study.

B Automation Bias

How we measured the existence of Automation Bias.

C Preliminary Results

What the study shows so far.

D Outlook

Where do we stand and where do we go from there?

A Participants

24 Participants

in Session 01

-0

loss

24 Participants 🥳

in Session 02

Distributions

of the demographic

Gender

	n_{abs}	n_{rel}
Male	10	41.67%
Female	14	58.33%
Other	0	0.00%

Age

	n_{abs}	n_{rel}
<20	1	4.17%
20 to 29	13	54.17%
30 to 39	2	8.33%
40 to 49	0	0.00%
50 to 59	4	16.67%
≥60	4	16.67%

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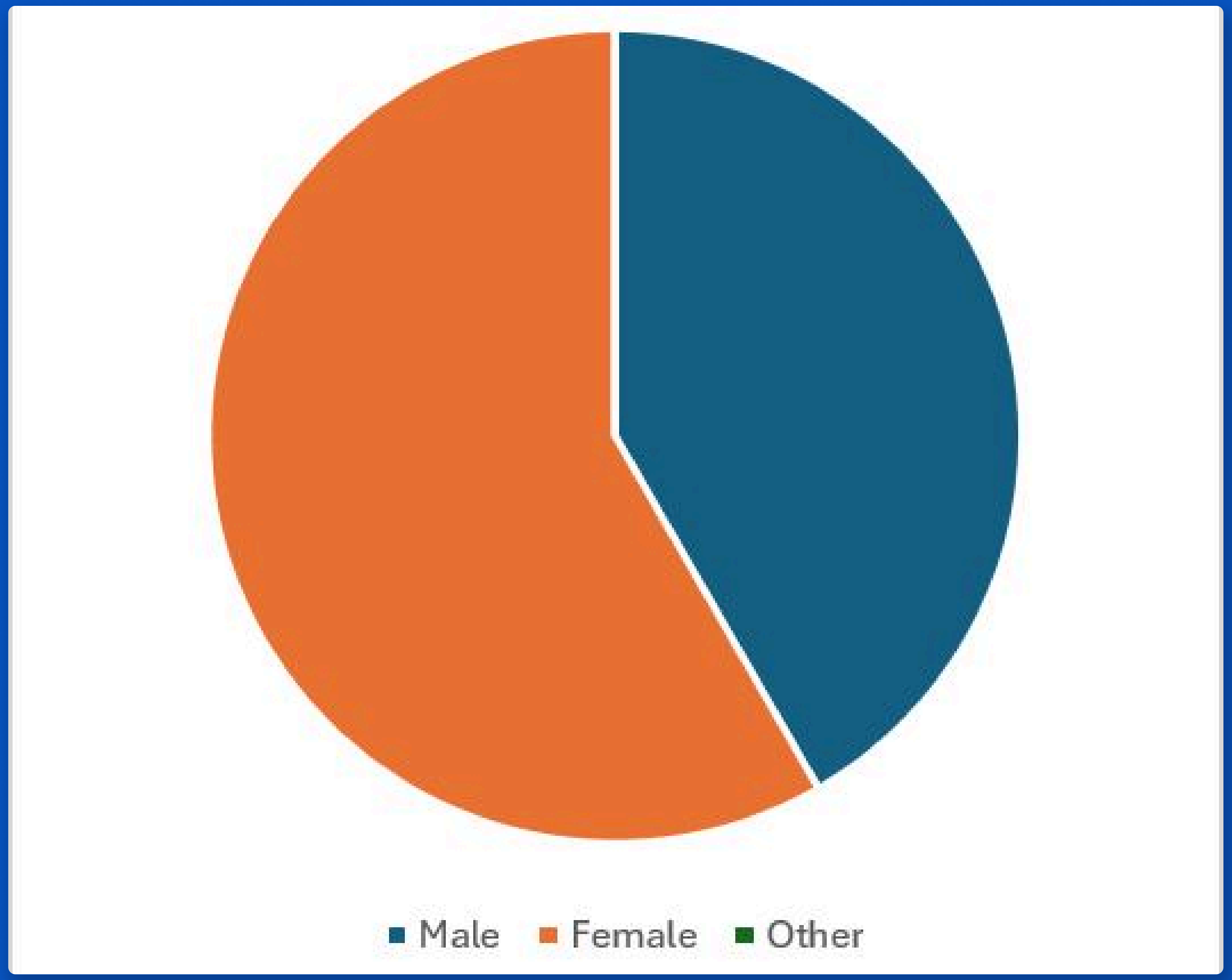
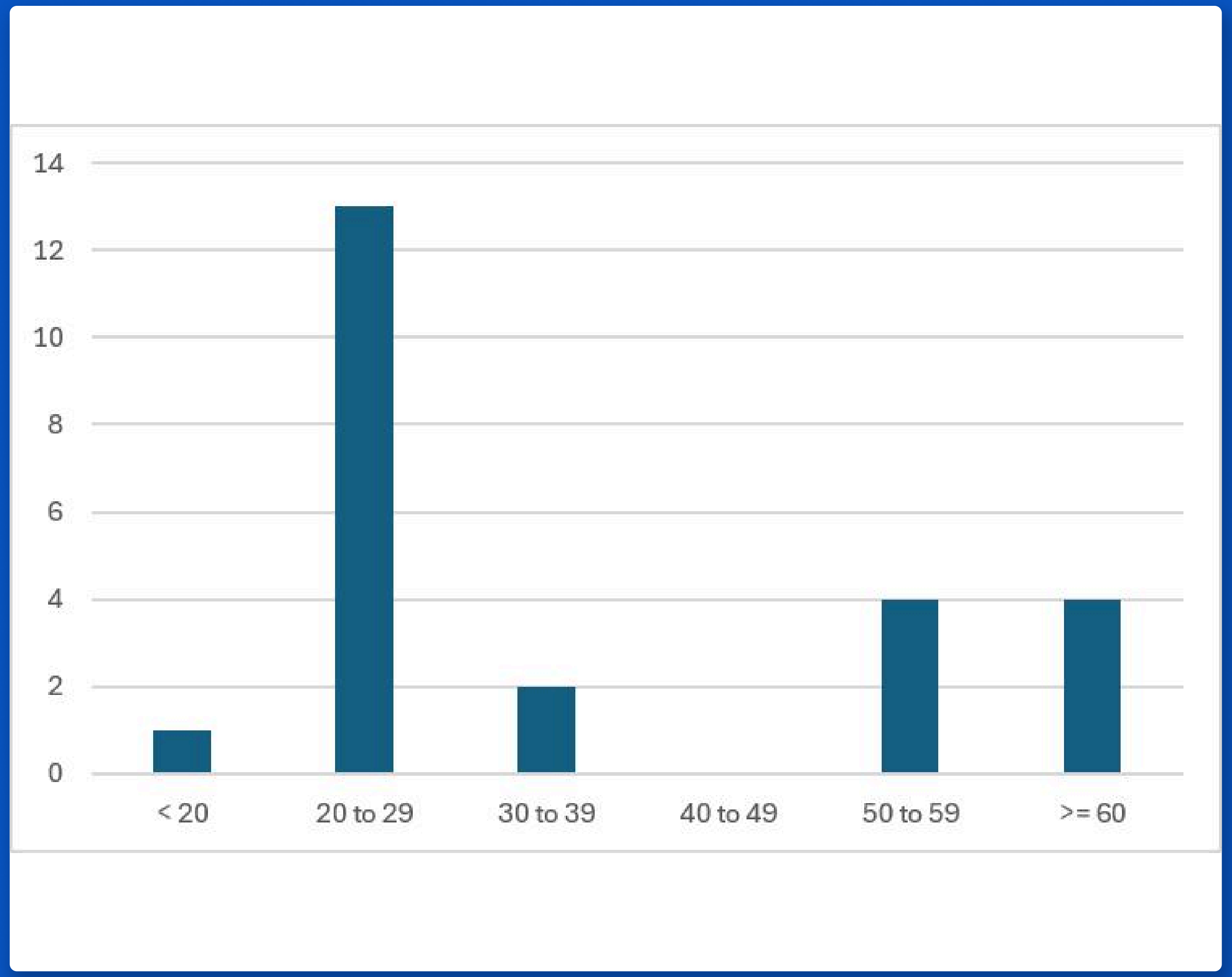




AI metadata

of the demographic

AI Usefulness (Gen.)

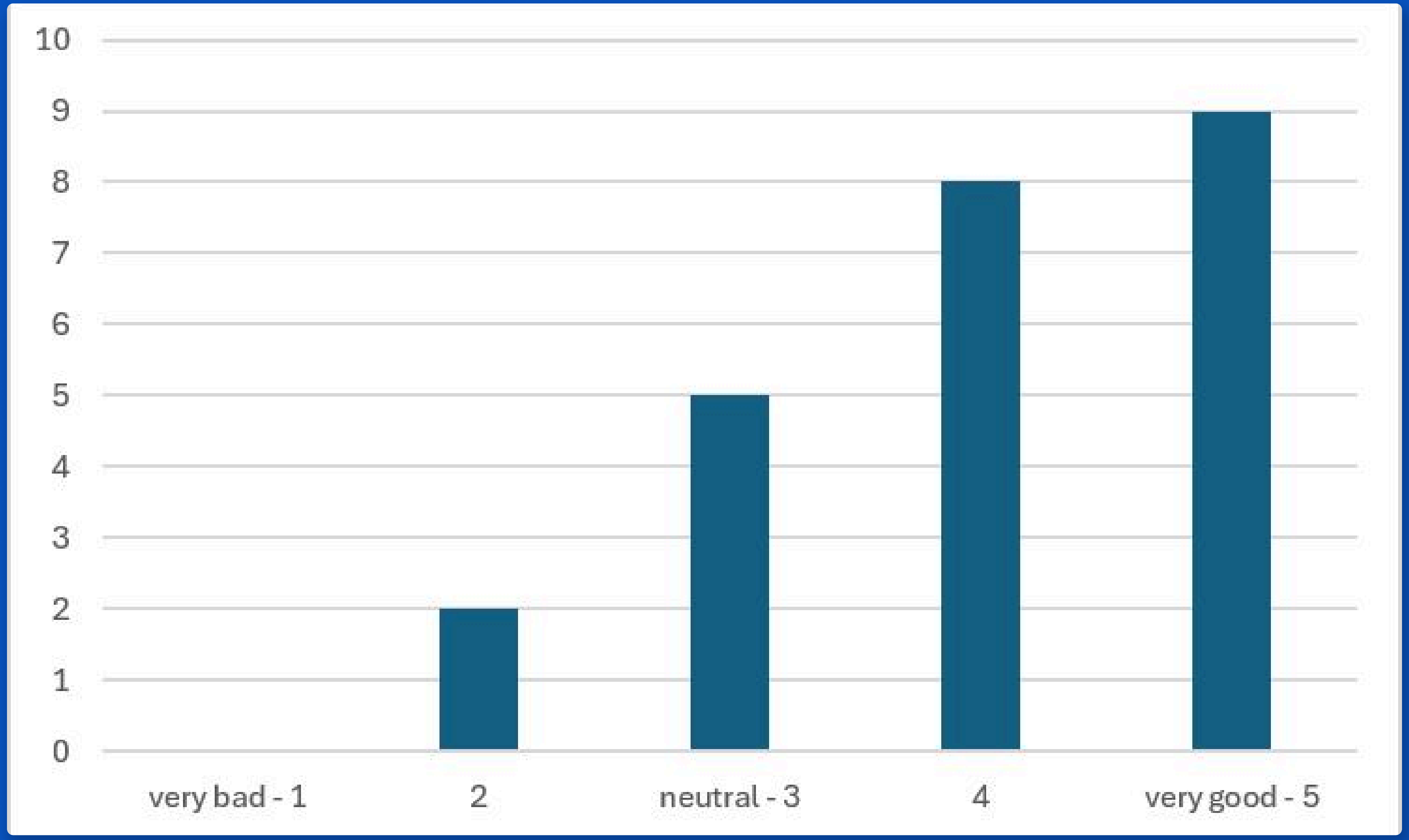
Rating	n_{abs}
very bad - 1	0
2	2
neutral - 3	5
4	8
very good - 5	9

AI Usefulness (Mental Health)

Rating	n_{abs}
very bad - 1	3
2	5
neutral - 3	7
4	6
very good - 5	3

AI Usefulness (Gen.)

Rating	n_{abs}
very bad - 1	0
2	2
neutral - 3	5
4	8
very good - 5	9



AI Usefulness (Mental Health)

Rating	n_{abs}
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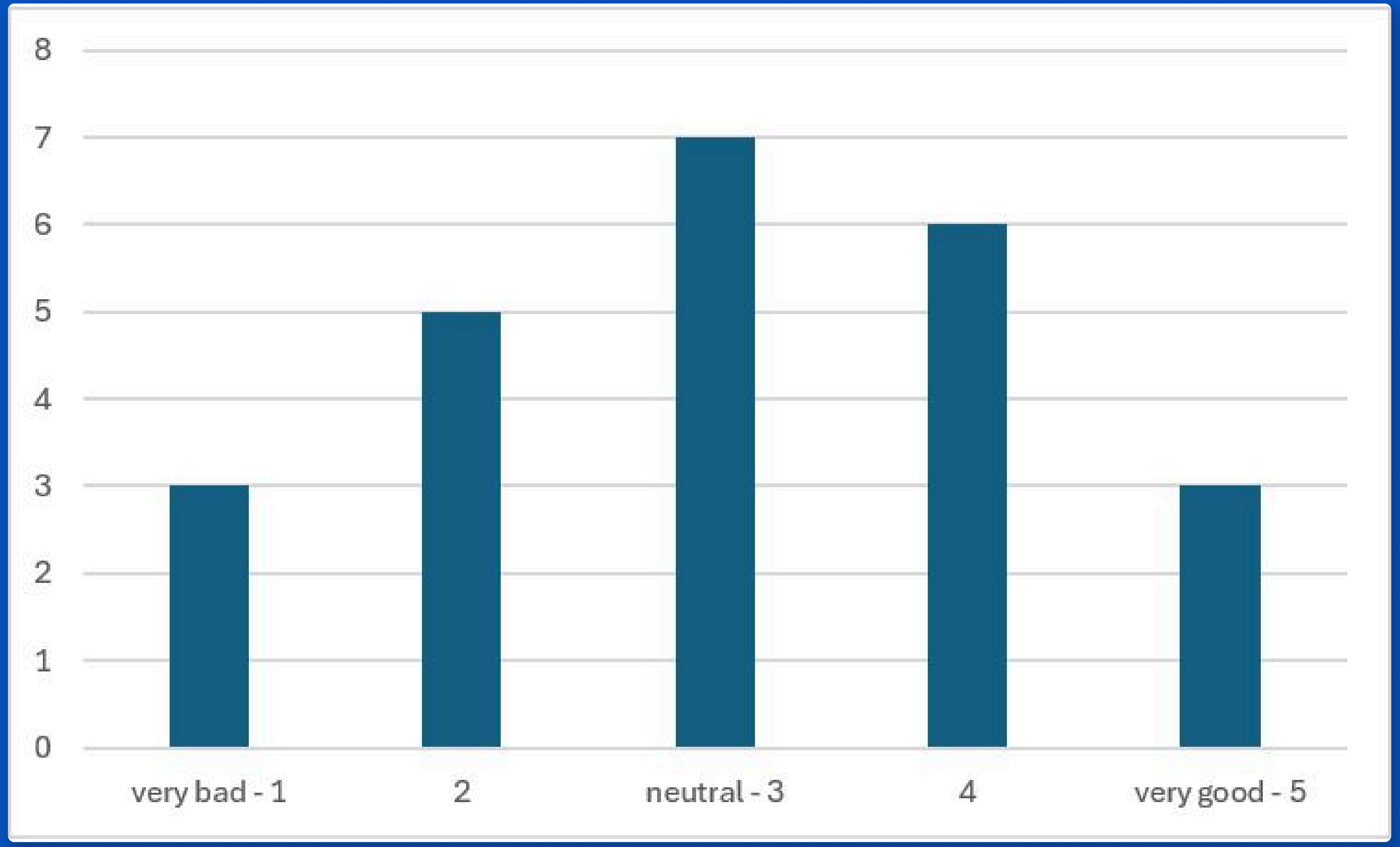

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Rating $r \in R_a := \{1, 2, 3, 4, 5\}$

Confidence $c \in [0, 100]$

Integrity $i \in T$

Automation Bias

$$B : (T, R_a, [0, 100]) \to [0, 1]$$

$$B(i, r, c) = (1 - i) \times \frac{r - 1}{5 - 1} \times \frac{c}{100}$$

Automation Bias

$$B(i, r, c) = (1 - i) \times \frac{r - 1}{5 - 1} \times \frac{c}{100}$$

Minimum Ack Automation Bias

B_{min} bei $i = 1, r = 3, c = 50$

$$\begin{aligned} B(0, 3, 50) &= (1 - 0) \times \frac{3 - 1}{4} \times \frac{50}{100} \\ &= 1 \times \frac{2}{4} \times 0.5 \\ &= 1 \times 0.5 \times 0.5 \\ &= 0.25 \\ B_{min} &> 0.25 \end{aligned}$$

Minimum acceptable Bias

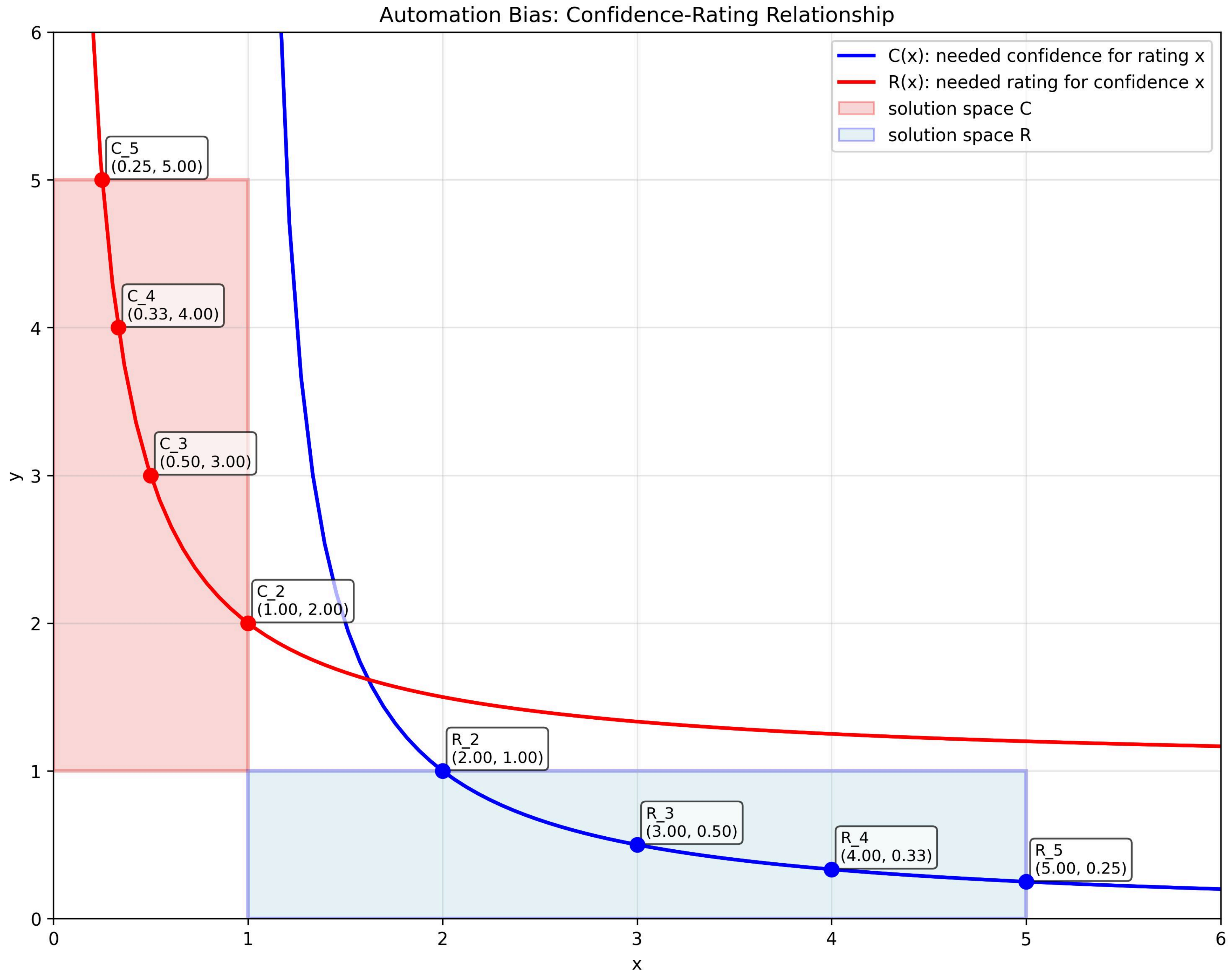
A graphical demonstration of which rating and confidence pairs are needed to satisfy our Automation Bias definition.

$$B_{min} = \frac{r-1}{4} \times c$$

$$0.25 = \frac{r-1}{4} \times c$$

$$\frac{1}{4} = \frac{(r-1) \times c}{4}$$

$$1 = (r-1) \times c$$



$$r = \frac{1}{c} + 1 \qquad c = \frac{1}{r-1}$$

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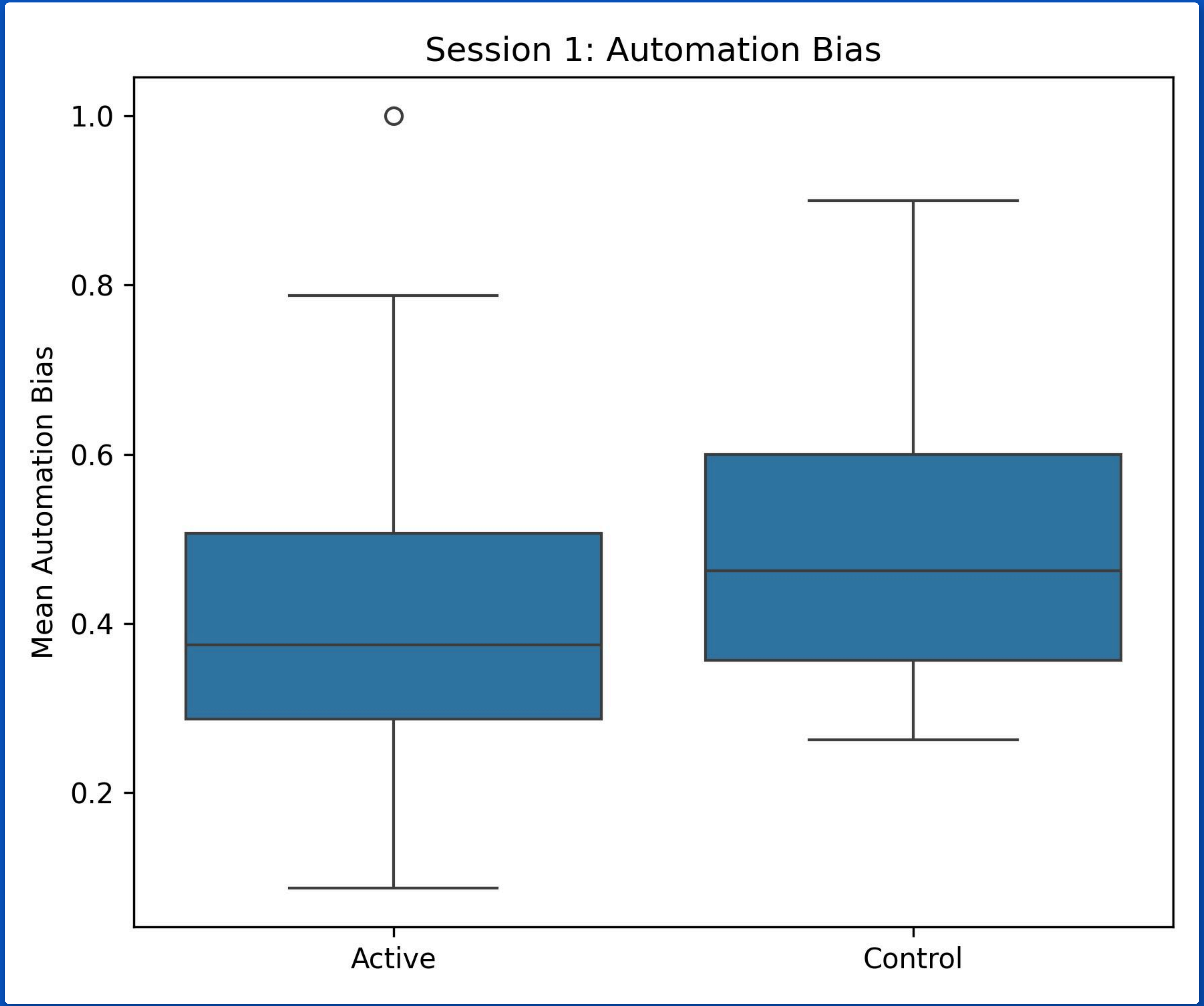
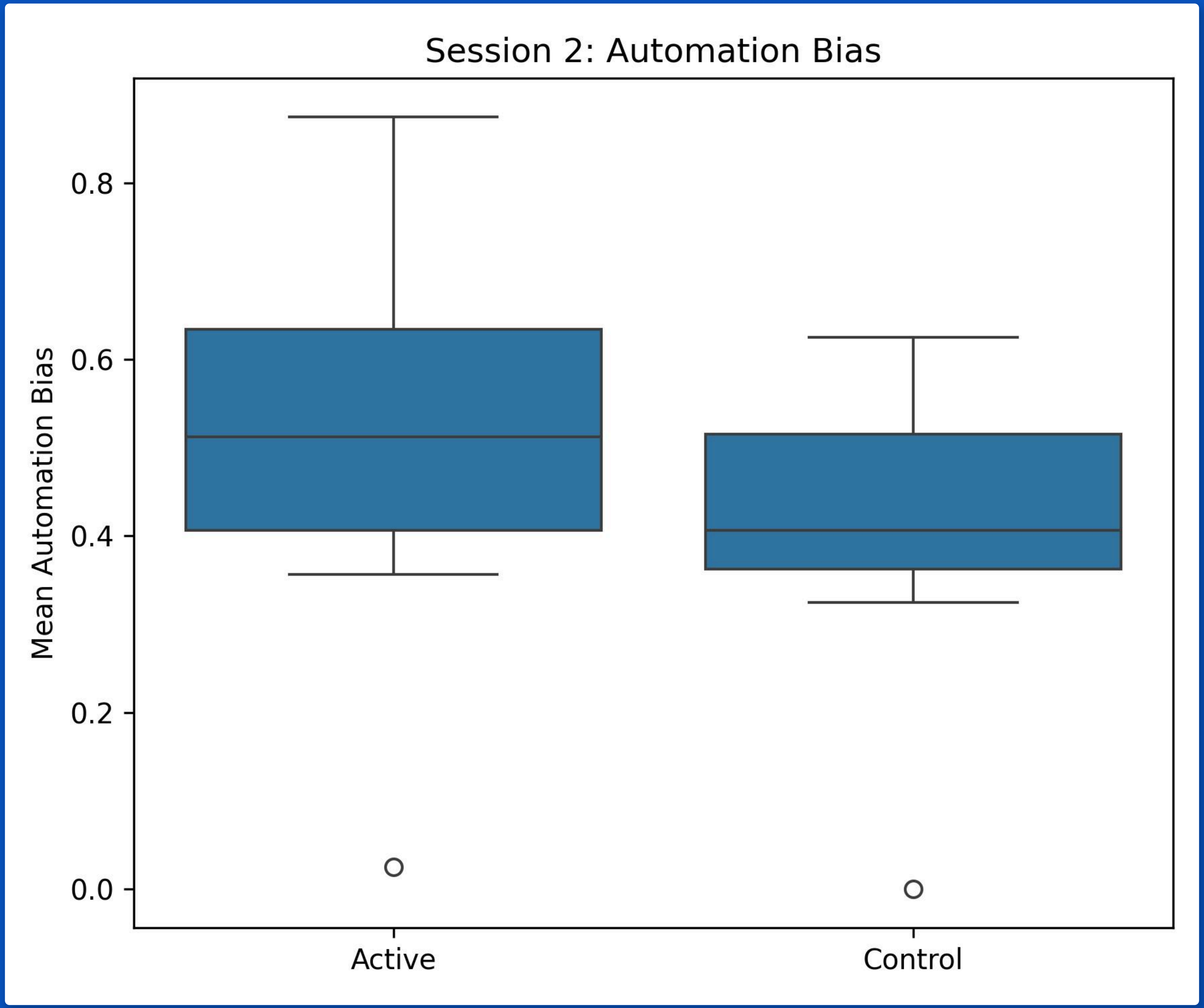
Comparison Active vs. Control Group Session 01

bias rating (1)	bias rating (2)	bias rating (3)	DISCLAIMER	bias_mean	session
0.0	0.225	0.0	1	0.1125	Session 1
0.0	0.0	0.675	1	0.3375	Session 1
0.45	0.0	0.5	0	0.475	Session 1
0.4	0.0	0.2	1	0.30000000000000004	Session 1
0.5249999999999999	0.0	0.0	0	0.26249999999999996	Session 1
0.07500000000000001	0.1	0.0	1	0.08750000000000001	Session 1
0.6000000000000001	0.0	0.6000000000000001	0	0.6000000000000001	Session 1
0.325	0.2	0.0	1	0.2625	Session 1
0.5249999999999999	0.0	0.4	0	0.46249999999999997	Session 1
0.0	0.5249999999999999	0.225	1	0.37499999999999994	Session 1
0.0	0.6375	0.6000000000000001	1	0.61875	Session 1
0.35	0.5	0.0	1	0.425	Session 1
0.0	0.5249999999999999	0.0	0	0.26249999999999996	Session 1
0.9	0.675	0.0	1	0.7875000000000001	Session 1
0.9	0.9	0.0	0	0.9	Session 1
1.0	1.0	0.0	1	1.0	Session 1
0.0	0.45	0.45	1	0.45	Session 1
0.75	0.75	0.0	0	0.75	Session 1
0.0	0.4	0.675	1	0.5375000000000001	Session 1
0.2125	0.7124999999999999	0.0	0	0.46249999999999997	Session 1
0.35	0.4	0.0	1	0.375	Session 1
0.35	0.2	0.0	1	0.275	Session 1
0.0	0.475	0.475	1	0.475	Session 1
0.2375	0.475	0.0	0	0.35624999999999996	Session 1

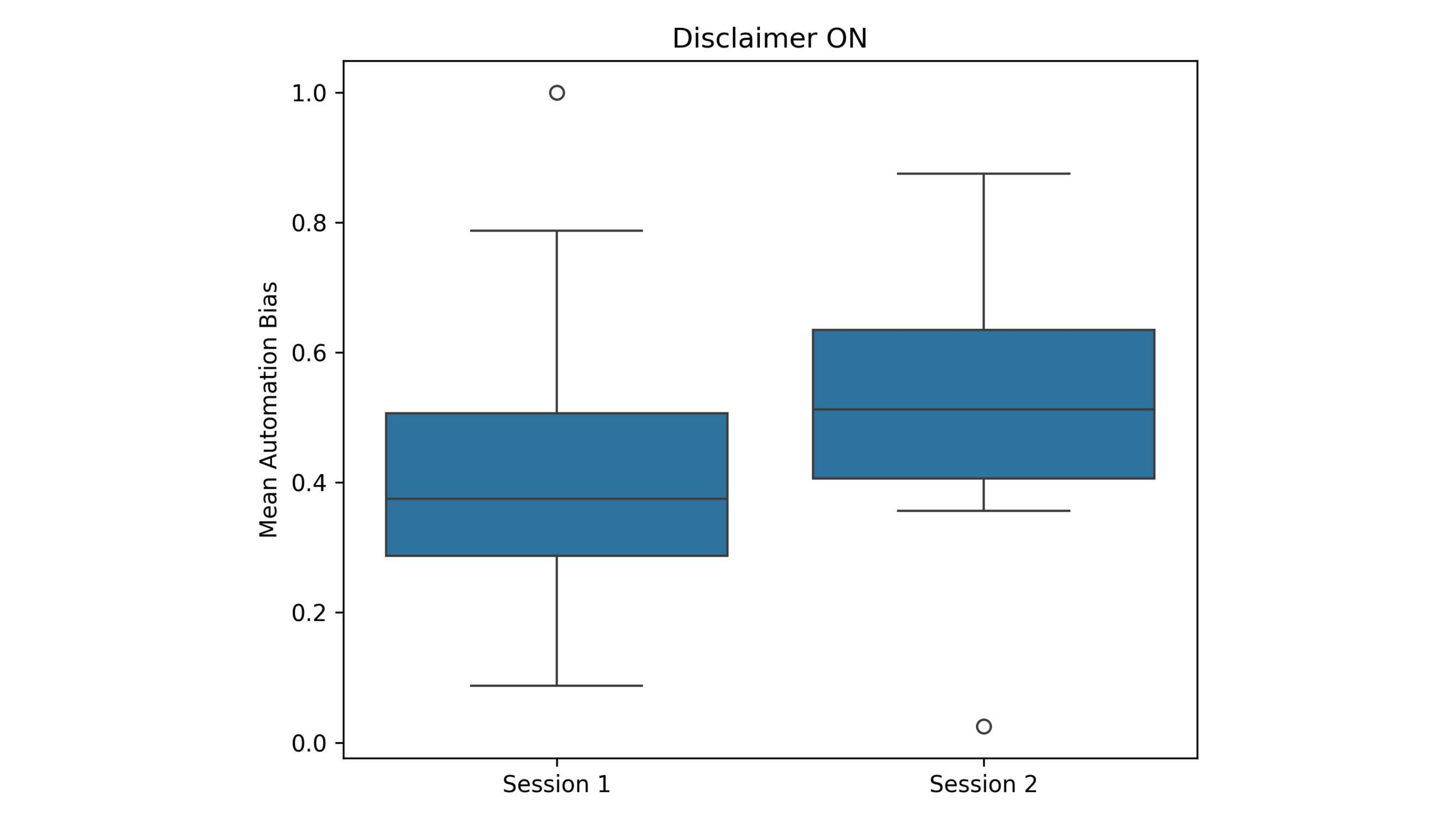


Comparison Active vs. Control Group Session 02

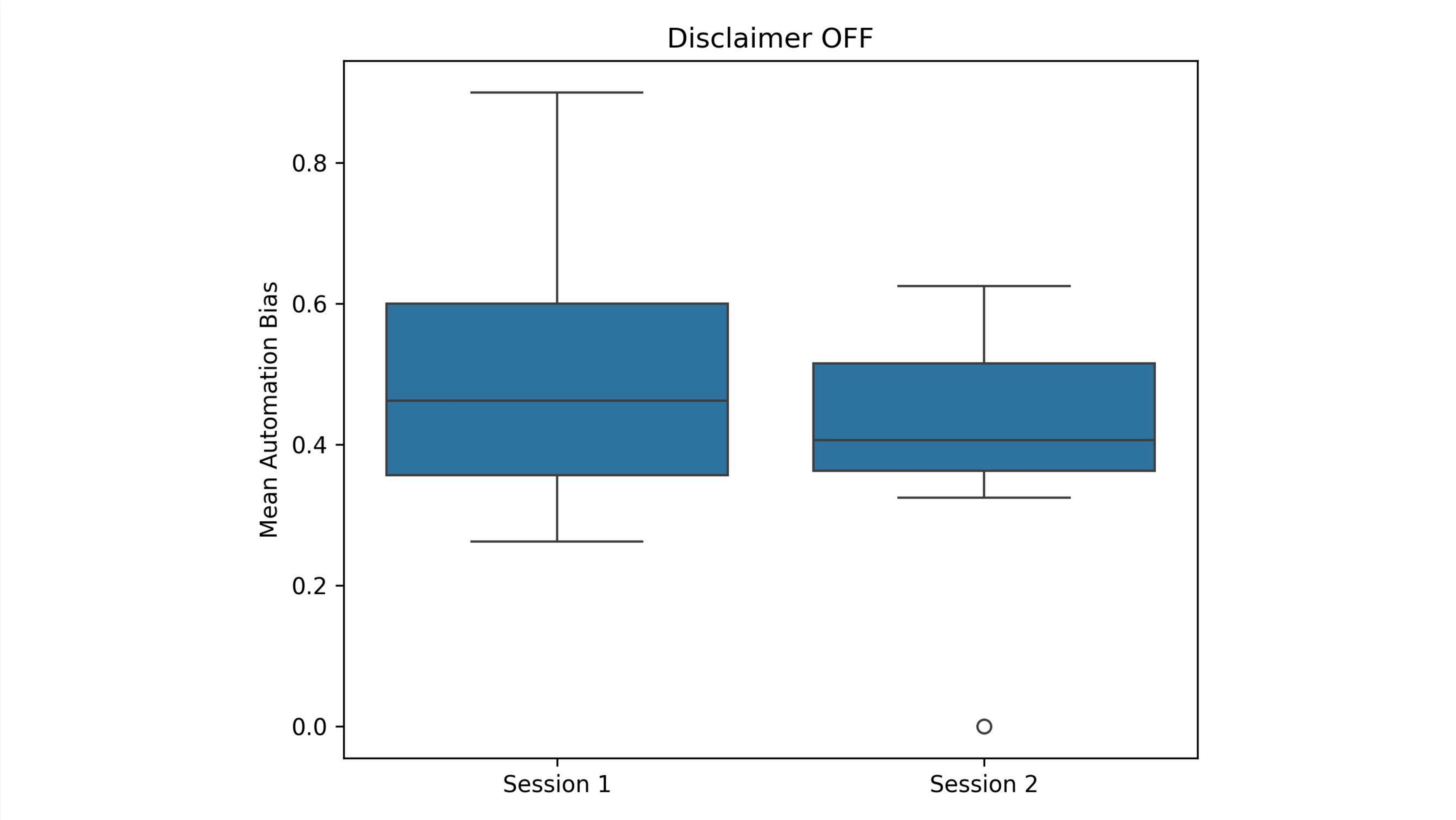
bias rating (1)	bias rating (2)	bias rating (3)	DISCLAIMER	bias_mean	session
0.0	0.0	0.75	0	0.375	Session 2
0.2	0.0	0.45	0	0.325	Session 2
0.0	0.125	0.125	0	0.125	Session 2
0.0	0.6000000000000001	0.225	0	0.41250000000000003	Session 2
0.0	0.675	0.7124999999999999	0	0.69375	Session 2
0.025	0.025	0.0	0	0.025	Session 2
0.225	0.0	0.0	0	0.1125	Session 2
0.5	0.25	0.0	0	0.375	Session 2
0.475	0.7124999999999999	0.0	0	0.59375	Session 2
0.75	0.0	0.5	0	0.625	Session 2
0.0	0.5249999999999999	0.44999999999999996	0	0.48749999999999993	Session 2
0.45	0.0	0.4	0	0.42500000000000004	Session 2
0.0	0.0	0.0	0	0.0	Session 2
0.75	0.0	0.25	0	0.5	Session 2
0.9	0.0	0.44999999999999996	0	0.675	Session 2
0.75	1.0	0.0	0	0.875	Session 2
0.4	0.0	0.4	0	0.4	Session 2
0.75	0.0	0.5	0	0.625	Session 2
0.675	0.6000000000000001	0.0	0	0.6375000000000001	Session 2
0.0	0.4	0.4	0	0.4	Session 2
0.0	0.2375	0.475	0	0.35624999999999996	Session 2
0.5249999999999999	0.0	0.5249999999999999	0	0.5249999999999999	Session 2
0.0	0.45	0.675	0	0.5625	Session 2



Comparison Session 01 vs. Session 02



Active



Control

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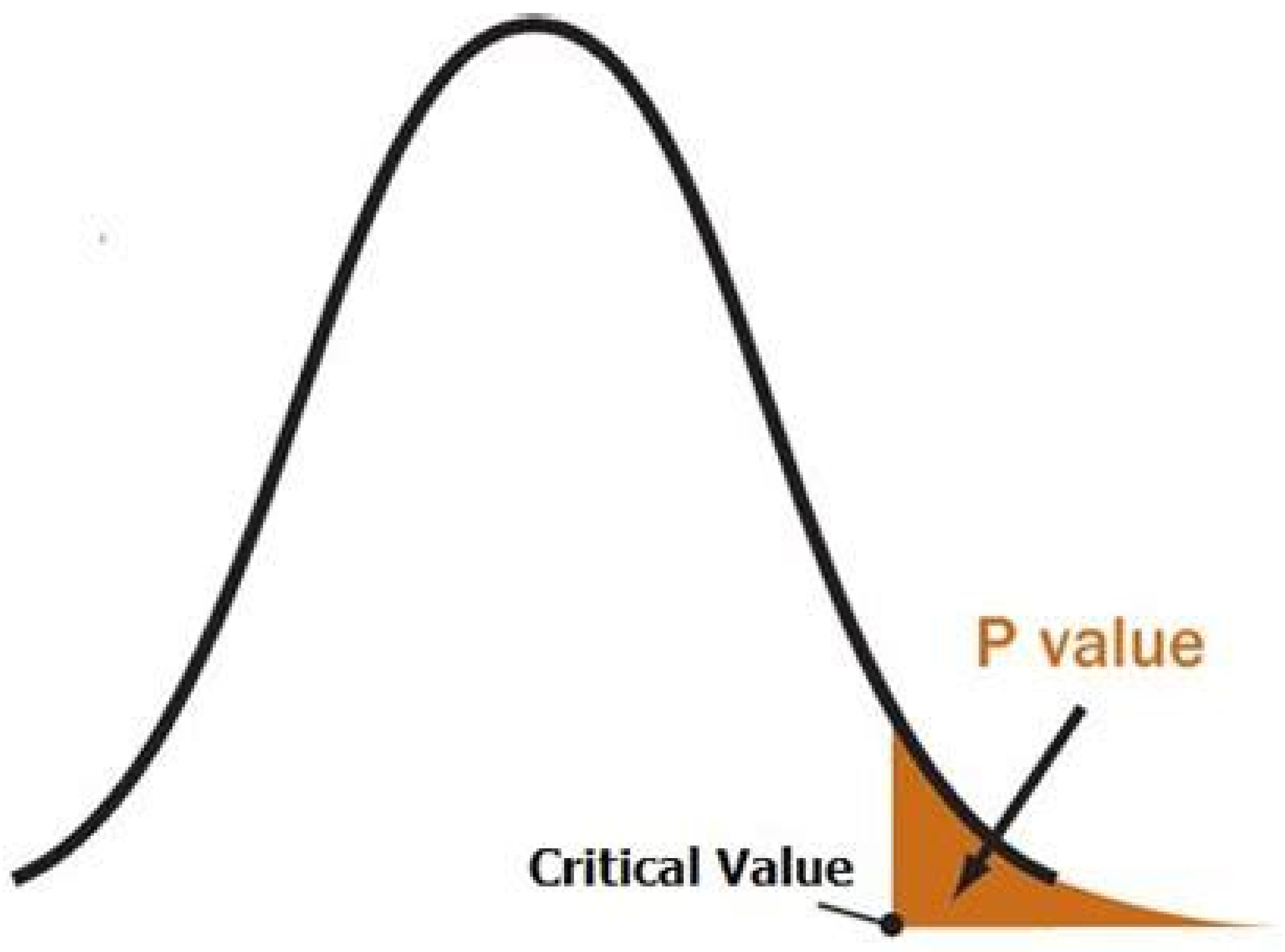
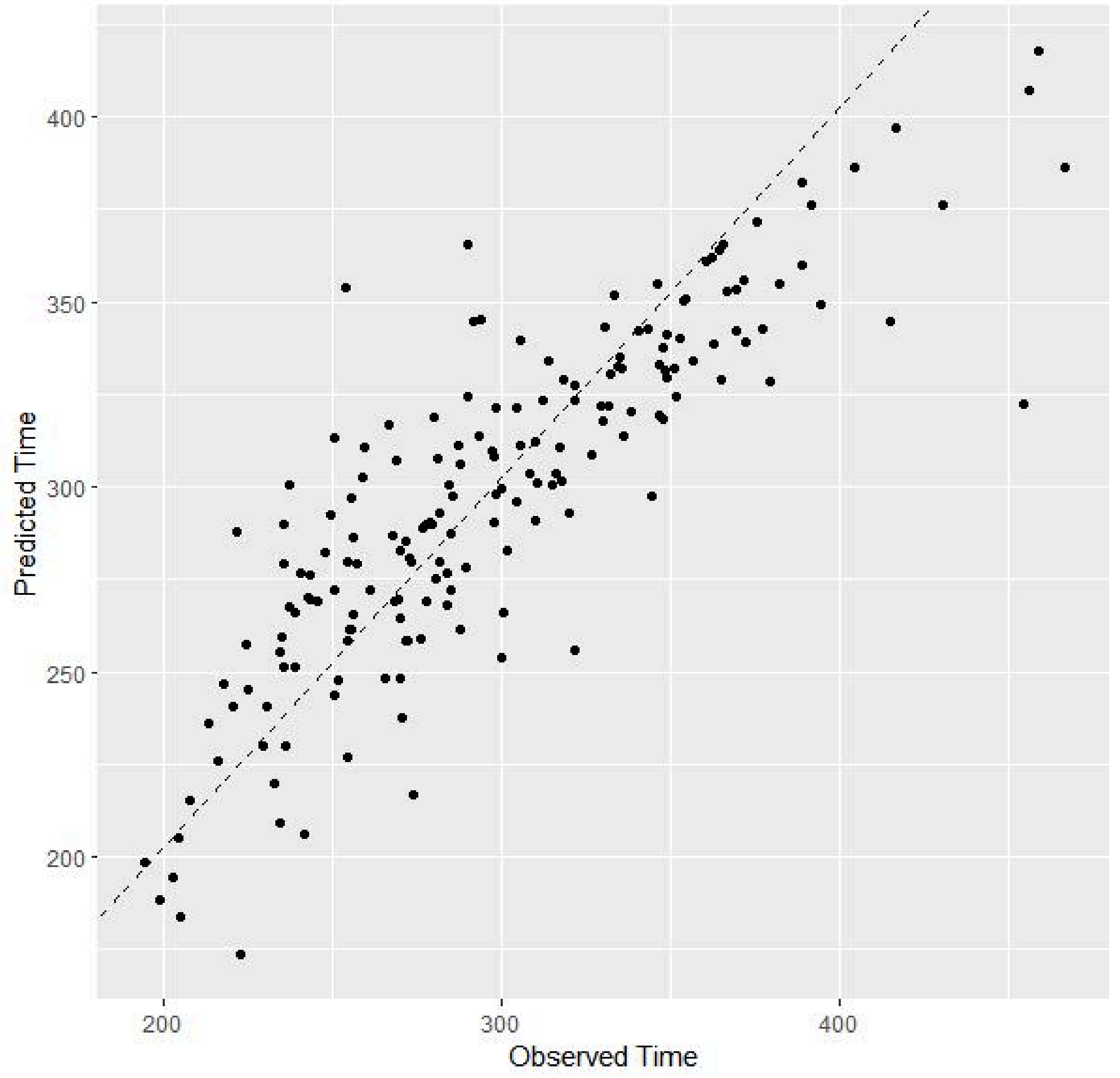
Sigificance Testing

To avoid uncertainty



Linear Mixed Model (LMM)

to avoid pseudo repetition



Implementation using statsmodels

Automation bias ~ condition * session + vignette + (1 | participant) $B \sim g \times s + c + (1|p)$

Fixed effects **Random effect**

Bias is modeled as a function of condition, session, their interaction, and vignette, with a random intercept for participant.

Thank you! 😍

You're awesome.